

Tutorial Sheet - 2

- ① Find a formula for the n th partial sum of the series

$$\frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \frac{1}{4 \cdot 5} + \dots + \frac{1}{(n+1)(n+2)} + \dots$$

and use it to find the series' sum if the series converges.

- ② Find the sum of the series

$$\sum_{n=0}^{\infty} \left(\frac{5}{2^n} + \frac{1}{3^n} \right)$$

- ③ Express the repeating decimal $0.\overline{d}$ as the ratio of two integers.

- ④ Use n th Term Test for divergence to show that the series is divergent.

i) $\sum_{n=1}^{\infty} \frac{n(n+1)}{(n+2)(n+3)}$

ii) $\sum_{n=0}^{\infty} \frac{e^n}{e^n + n}$

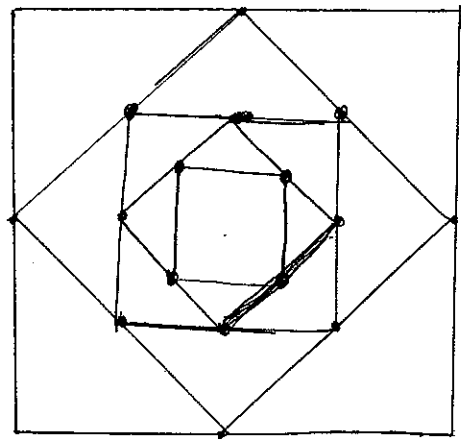
- ⑤ Find the sum of the series $\sum_{n=1}^{\infty} \frac{6}{(2n-1)(2n+1)}$

- ⑥ Which of the following series converge, and which diverge?

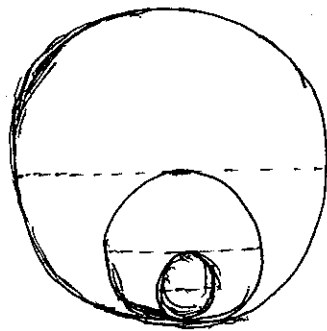
i) $\sum_{n=0}^{\infty} \frac{n!}{100^n}$

ii) $\sum_{n=1}^{\infty} \frac{2^n + 4^n}{3^n + 4^n}$

- ⑦ Can you make an infinite series of nonzero terms that converges to any number you want? Explain.
- ⑧ Show by example that $\sum \frac{a_n}{b_n}$ may diverge even though $\sum a_n$ and $\sum b_n$ converge and no b_n equals 0.
- ⑨ Find convergent geometric series $A = \sum a_n$ and $B = \sum b_n$ that illustrate the fact that $\sum a_n b_n$ may converge without being equal to AB .
- ⑩ Show by example that $\sum \frac{a_n}{b_n}$ may converge to something other than $\frac{A}{B}$ even when $A = \sum a_n$, $B = \sum b_n \neq 0$ and no b_n equals 0.
- ⑪ If $\sum a_n$ converges and $a_n > 0$ for all n , can anything be said about $\sum (\frac{1}{a_n})$? Give reasons for your answer.
- ⑫ The accompanying figure shows the first five of a sequence of squares. The outermost square has an area of 4 m^2 . Each of the other squares is obtained by joining the midpoints of the edges of the squares before it. Find the sum of the areas of all the squares.



- (13) The largest circle in the accompanying figure has radius 1. Consider the sequence of circles of maximum area inscribed in semi-circles of diminishing size. What is the sum of the areas of all of circles?



- (14) Use Integral Test to determine if the series converge or diverge. Be sure to check that the conditions of the Integral Test are satisfied.

i) $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$

ii) $\sum_{n=2}^{\infty} \frac{\ln(n^2)}{n}$

iii) $\sum_{n=2}^{\infty} \frac{1}{n(1 + \ln^2 n)}$

iv) $\sum_{n=1}^{\infty} \frac{\ln n}{n^2}$

(15) Is it true that if $\sum_{n=1}^{\infty} a_n$ is a divergent series of positive numbers, then there is also a divergent series $\sum_{n=1}^{\infty} b_n$ of positive numbers with $b_n < a_n$ for every n ? Is there a "smallest" divergent series of positive numbers?

(16) Use the Cauchy condensation test to show that $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^p}$ converges if $p > 1$ and diverges if $p \leq 1$.

(17) (a) For the series $\sum_{n=1}^{\infty} \frac{1}{n^3}$, use the inequality

$$s_n + \int_{n+1}^{\infty} f(x) dx \leq S \leq s_n + \int_n^{\infty} f(x) dx$$

with $n=10$ to find an interval containing the sum S .

(b) Use the midpoint of the interval found in part (a) to approximate the sum of the series. What is the maximum error for your approximation?

(18) Use the Direct Comparison Test to determine if each series converges or diverges.

i) $\sum_{n=1}^{\infty} \sqrt{\frac{n+4}{n^2+4}}$ ii) $\sum_{n=1}^{\infty} \frac{\cos^n n}{n^{3/2}}$ iii) $\sum_{n=1}^{\infty} \frac{3}{n+\sqrt{n}}$

(19) Use the Limit Comparison Test to determine if each series converges or diverges.

i) $\sum_{n=1}^{\infty} \frac{(n+1)}{n(n+1)(n+2)}$ ii) $\sum_{n=1}^{\infty} \frac{\sqrt{n+1}-1}{(n+2)^3-1}$ iii) $\sum_{n=1}^{\infty} \frac{1}{1^2+2^2+\dots+n^2}$